CTPIP

Dataset Information

Document control	
Document ID	CTPIP-DI-001
Project	Clinical Trial Protocol Intelligence Platform (CTPIP)
Version	1.1
Date	21 October 2026
Status	Released

Version	Date	Description of change
1.1	21 October 2026	Initial formal release.

Summary

- 187 unique breast-cancer clinical trial protocol documents.
- Composition, sources, topics and quality notes.
- Location of the data and the catalog file.
- How to add further documents.

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1. Dataset at a Glance

- The dataset is a collection of clinical trial protocol documents (PDF) about breast cancer.
- A protocol is the master plan of a medical study: purpose, eligible patients, treatments and doses, safety monitoring, measurements and analysis.
- Sources: pharmaceutical companies, hospitals, universities and cooperative groups; written between 2006 and 2025.

Measure	Value
PDF files in the project folders	190 (in five numbered folders)
Unique documents after duplicate removal	187 (3 files are exact copies)
Total pages	19,413 (smallest 5, largest 473, average about 104)
Total size	about 443 MB of PDFs
Protocols and statistical analysis plans	168 protocols and 19 SAPs
Documents with a registry (NCT) number	93 documents, 90 distinct trials
Text type	174 text, 12 mixed, 1 scanned (read with OCR)
Processing result	187 processed, 0 failed

2. Location of the Data

Location	Content
`1-19/`, `20-38/`, `39-56/`, `57-100/`, `101-185/`	Original PDFs; folder name = range of document numbers (19 + 20 + 18 + 44 + 89 = 190 files).
`data/pdfs/`	Copy kept by CTPIP for each unique document (name d plus 12 characters). Used to display pages.
`data/artifacts/`	Per-document checkpoint files.
`data/ctpip.sqlite3`	Metadata database: documents, sections, passages, tables, warnings, history.
Neo4j database `ctpip`	The knowledge graph.
`docs/Dataset_Catalog.csv`	One row per document (section 7).

- The original folders are the source of truth: if `data/` or the graph is lost, Load dataset folder rebuilds everything.
- On a new computer the dataset is loaded automatically at the first start (about 30-60 minutes); `data/` is not part of the repository.

3. File Naming

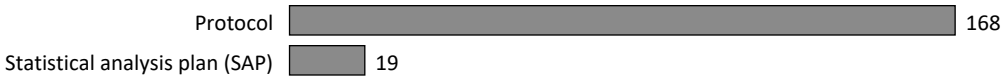
- Pattern: number, dot, short descriptive name. Examples:

```
7. ASCENT.pdf
101. Neoadjuvant carboplatin + paclitaxel +- atezolizumab in TNBC.pdf
110. BRAVO - niraparib vs physician's choice in gBRCAmut HER2- ABC (Protocol).pdf
110. BRAVO - niraparib vs physician's choice in gBRCAmut HER2- ABC (SAP).pdf
```

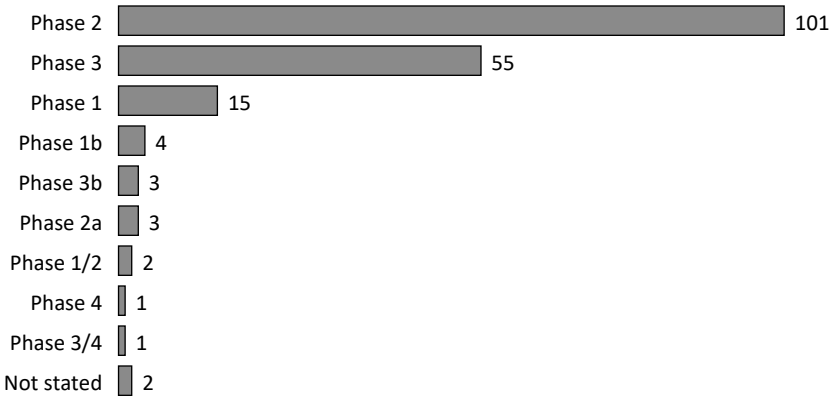
- The number is only an ordering label.
- Names often contain the trial acronym, drugs and disease setting.
- A suffix (Protocol) or (SAP) distinguishes several files of one study.
- Some names are only labels (for example "59. Neoadjuvant.pdf"); the real title is read from inside the PDF.

4. Composition

4.1 Document type

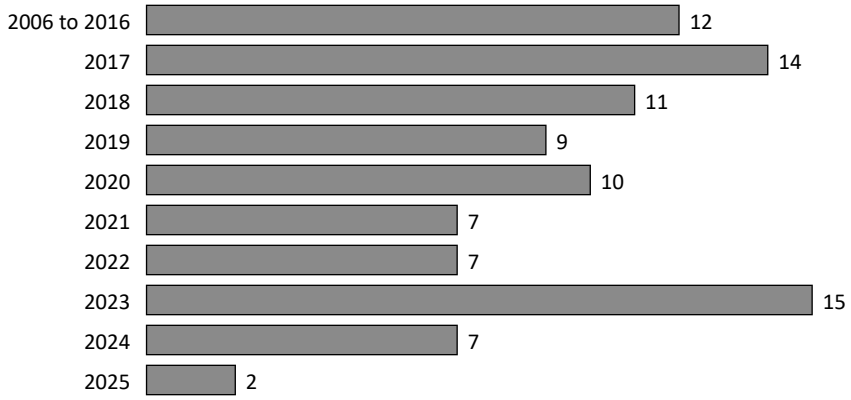


4.2 Study phase



- Phase 1: safety and dose in small groups.
- Phase 2: signs of effectiveness.
- Phase 3: comparison with standard treatment in large groups.
- Phase 4: follow-up of approved drugs.

4.3 Version date printed in the document (94 files show none)



4.4 Document length



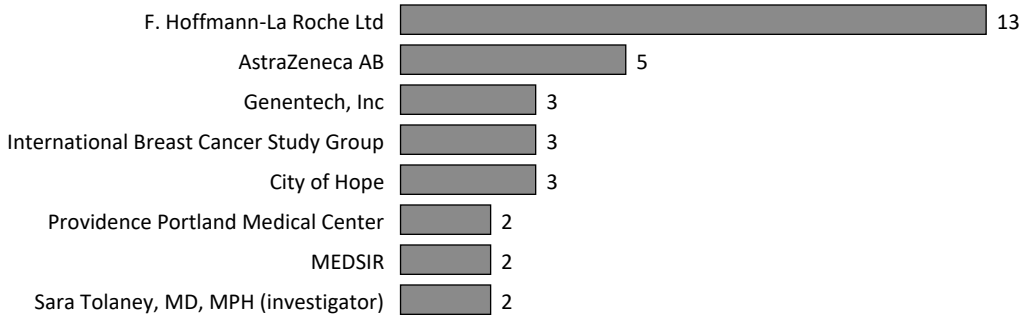
4.5 Folders

Folder	Files / unique	Pages	Types
`1-19/`	19 / 18	1,424	11 protocols, 7 SAPs
`20-38/`	20 / 20	3,034	20 protocols
`39-56/`	18 / 17	2,042	17 protocols
`57-100/`	44 / 43	5,288	43 protocols
`101-185/`	89 / 89	7,625	77 protocols, 12 SAPs

- Identical file pairs: #10 and #11, #49 and #50, #58 and #59. Duplicates are detected by content, not by name.

5. Sponsors

- Sponsor names are read from the title pages; 94 documents state no sponsor in a readable form (cells are empty in the catalog).
- Most frequent sponsors:



- The collection mixes industry-sponsored global trials, academic hospital trials and cooperative-group trials (NSABP, ECOG-ACRIN, SWOG, IBCSG).

6. Topics Covered

Theme	Examples in the data
HER2-positive disease	trastuzumab, pertuzumab, T-DM1 (KATHERINE), T-DXd (DESTINY-Breast), tucatinib (HER2CLIMB)
Hormone-receptor-positive disease	palbociclib, ribociclib, abemaciclib, fulvestrant, letrozole, everolimus, alpelisib
Triple-negative disease	pembrolizumab (KEYNOTE-355), atezolizumab, sacituzumab govitecan (ASCENT), carboplatin, PARP inhibitors
Settings	neoadjuvant, adjuvant and metastatic (TAILORx, NSABP B-47, ALEXANDRA, SOLE)
Supportive care and prevention	biosimilar filgrastim, rifaximin, elinzanetant, carvedilol, metformin, omega-3, letrozole prevention
Early phase and novel agents	first-in-human studies, vaccines, TIL therapy, ARV-471, basket trials

7. Catalog File

- `docs/Dataset_Catalog.csv` has one row per document and opens in Excel (UTF-8 with header row).
- Columns:

Column	Description
Doc No, Folder, File name	Number, containing folder and name of the PDF.
Study title	Title read from the PDF (first 220 characters); amendments may start with a summary of changes.
Document type	protocol or sap.
Trial number (NCT)	Registry ID found in the text (empty if none).
Sponsor protocol code	Sponsor's own code.

Column	Description
Phase, Sponsor, Version, Amendment, Version date	Taken from the title page when stated.
Pages, Size (MB)	Length and file size.
PDF kind	text, mixed or scanned.
Entities found	Distinct items stored for the document.
Relationships in graph	Facts written to the graph for the document.
Warnings	Number of reading-quality notes.

8. Output of Processing

Output	Count	Meaning
Sections	25,727	Headings detected (about 138 per document).
Evidence chunks	95,181	Passages with page and position, including table rows.
Tables	4,821	Tables kept as rows and columns.
Entity mentions	305,219	Known items spotted in text (350 different items).
Clinical relationships	29,047	Facts with claim, page, quote; 28,716 validated.
All relationships	175,589	Including structure, terminology and mentions.
Protocol nodes	187	One per document.
Trials	133	Registered trials (a trial may have several documents).
Shared entities	693	218 drugs, 57 adverse events, 35 biomarkers, 28 endpoints, 23 conditions.

9. Data Quality

- Every reading problem is recorded as a warning; there are 168 warnings of severity warning and no errors.

Warning	Count	Meaning / action
`table_quality`	1,059	A table was hard to read; check the page image if it matters.
`ocr_used`	195	Page read by OCR; small spelling errors possible.
`letter_spaced_text`	97	Text stored with spaces between letters; matching may suffer.
`nct_not_found`	94	No registry number printed; trial cannot be linked by NCT.

Warning	Count	Meaning / action
`empty_section`	42	Heading without captured text.
`sections_mostly_unknown`	14	More than half of the headings not classified.
`eligibility_not_found`	10	No inclusion/exclusion section (normal for some SAPs and amendments).
`no_headings`	3	Almost no headings; boundaries unreliable.

10. Limitations

- Breast cancer only: the dictionary is tuned to breast-oncology drugs, biomarkers and endpoints.
- Duplicates: three identical pairs give 187 documents from 190 files.
- Mixed document kinds: protocols, SAPs and amendment summaries are linked to one trial when they share an NCT number.
- Sponsor and NCT coverage is partial: only what is printed in the PDF can be found.
- Not a clinical decision tool: CTPIP organises what documents state.
- No patient data: protocols describe studies, not individual patients.

11. Adding Documents

1. Check that the file is a normal PDF (not password-protected) under 200 MB.
2. Open Protocols and drag the file in or click Choose PDFs; for many files copy them into the project folder and click Load dataset folder.
3. Wait until the status is ready.
4. Open Data Quality to review any new look-alike drug names.
5. To practise, use `docs/demo/Sample_Protocol_SAMPLE-PEMBRO-001.pdf` (made-up) and delete it afterwards if it is not wanted.